

CUDA Multi CPU Comparison Assignment

Student Information

Field	Details
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Course	Parallel Computing / CUDA Programming
Technology	CUDA C++

Objective

The objective of this assignment is to implement a CUDA-based GPU program that compares two floating-point vectors received from two CPU-based Python programs.

For every element:

- If $a[i] > b[i] \rightarrow \text{output } 1.0$
- If $a[i] == b[i] \rightarrow \text{output } 0.0$
- If $a[i] < b[i] \rightarrow \text{output } -1.0$

The CUDA kernel performs parallel comparison of vector elements and stores the result into output CSV files.

Software Requirements

- NVIDIA CUDA Toolkit
- GCC Compiler
- Linux Environment
- Python 3
- CUDA-enabled GPU

Files Used

File Name	Purpose
input_a.csv	Input vector A
input_b.csv	Input vector B
output_a.csv	CUDA comparison output
output_b.csv	CUDA comparison output
results.csv	Submission file
multi_cpu.cu	CUDA source code

CUDA Kernel Logic

The CUDA kernel compares vector elements using branchless comparison:

```
c[i] = (a[i] > b[i]) - (a[i] < b[i]);
```